2025-2026 春《数据结构与算法》第六次作业题解

助教-陈孙一硕

A 归并排序兼求逆序对数量

归并排序求逆序对。但事实上我感觉不太好用，有兴趣的可以去学习树状数组求逆序对。

```
#include <bits/stdc++.h>
using namespace std;
#define int long long
int q[105], temp[105], cnt = 0;
void merge_sort(int l, int r) {
    if (l >= r) return ;
    int mid = (l + r) / 2, i = l, j = mid + 1, k = 0;
    merge_sort(l, mid);
    merge_sort(mid + 1, r);
    while (i <= mid && j <= r) {
        if (q[i] <= q[j]) temp[k++] = q[i++];
        else temp[k++] = q[j++], cnt += mid - i + 1;
    }
    while (i <= mid) temp[k++] = q[i++];
    while (j <= r) temp[k++] = q[j++];
    for (i = l, j = 0; i <= r; i++, j++) q[i] = temp[j];
}
signed main() {
    int n; cin >> n;
    for (int i = 0; i < n; i++) cin >> q[i];
    merge_sort(0, n - 1);
    cout << "count = " << cnt << endl;
    for (int i = 0; i < n; i++) cout<< q[i] << " ";
    return 0;
```

```
}
```

B 成绩排名

封装结构体，重载运算符，感觉有手就行。

```
#include<bits/stdc++.h>
using namespace std;
struct node{
    int ch,math,id;
    bool operator < (const node n1) const{
        if (n1.ch==ch&&n1.math==math) return id<n1.id;
        else if (n1.ch==ch) return n1.math<math;
        return n1.ch<ch;
    }
};
const int N=2e6+101;
node a[N];
int main(){
    int n; cin>>n;
    for (int i=1;i<=n;i++) cin>>a[i].id>>a[i].ch>>a[i].math;
    sort(a+1,a+n+1);
    for (int i=1;i<=n;i++) cout<<a[i].id<<endl;
    return 0;
}
```

C 基数排序

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;
int getDigit(int num, int d) {
    for (int i = 0; i < d; ++i) num /= 10;
    return num % 10;
```

```
}
void radixSort(vector<int>& arr, int maxDigits) {
    vector<int> tmp(arr.size());
    for (int d = 0; d < maxDigits; ++d) {
        int cnt[10] = {0};
        for (int x : arr) cnt[getDigit(x, d)]++;
        for (int i = 1; i < 10; ++i) cnt[i] += cnt[i-1];
        for (int i = arr.size()-1; i >= 0; --i) {
            int digit = getDigit(arr[i], d);
            tmp[--cnt[digit]] = arr[i];
        }
        arr = tmp;
        for (size_t i = 0; i < arr.size(); ++i) {
            cout << arr[i] << " ";
        }
        cout << endl;
        if (d == 2) break;
    }
}

int main() {
    int n;
    cin >> n;
    vector<int> arr(n);
    int maxVal = 0;
    for (int i = 0; i < n; ++i) {
        cin >> arr[i];
        if (arr[i] > maxVal) maxVal = arr[i];
    }
    int maxDigits = 0;
    while (maxVal > 0) {
        maxDigits++;
        maxVal /= 10;
    }
    radixSort(arr, maxDigits);
    return 0;
```

```
}
```